

Ultra-low power self-interconnected multi-point touch IC

V1. 0

1. Essentials

The CST9220 ultra-low-power self-integrated multi-point touch chip supports single-layer, multi-layer modules, and various touch patterns. It employs multi-channel, variable-frequency, and high-voltage driving schemes, offering a higher signal-to-noise ratio and superior anti-interference performance compared to traditional single-channel, single-frequency, low-voltage driving solutions. Additionally, the chip's built-in self-integrated capacitive sensing module, combined with an intelligent scanning algorithm, enables rapid response while delivering excellent noise rejection, waterproofing, and low-power consumption characteristics.

2. Characteristic

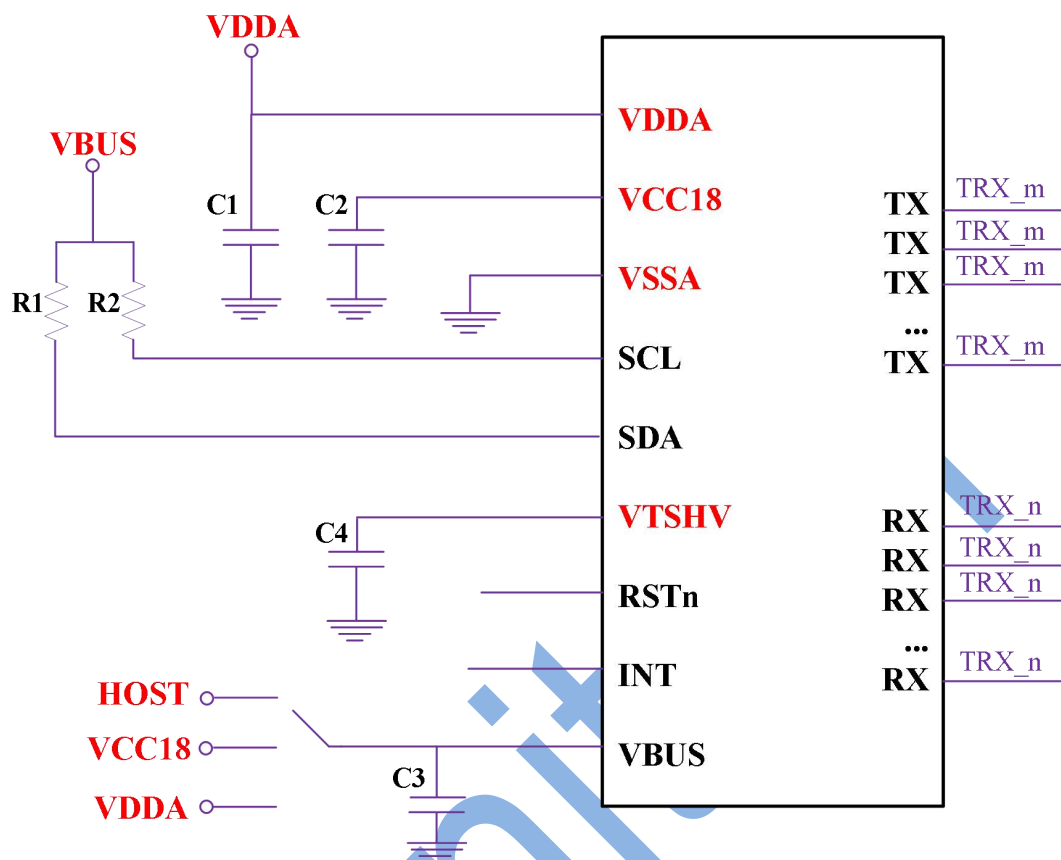
- High-performance capacitor detection circuit and DSP module
 - Equipped with a built-in 32-bit processor, featuring dual clock speeds of 48 MHz and 32 kHz, ensuring both compatibility and power efficiency;
 - The self-interlocking detection module provides excellent waterproof performance;
 - Supports simultaneous drive of multiple high-voltage channels, featuring high sensitivity and a high signal-to-noise ratio;
 - Dynamic wide-range frequency hopping technology provides enhanced anti-interference capability;
 - Tracks environmental changes and performs automatic calibration;
 - Supports standby gestures, as well as custom gestures such as single tap, double tap, left/right swipe, up/down swipe, and 'O', 'Z', 'C', '3';
 - Supports online programming;
- performance index
 - Typical refresh rate: 100Hz;
 - Supports up to 5-finger touch;
 - Dynamic mode typical power consumption: 1.4 mA (100 Hz);
 - Typical power consumption in monitoring mode: 12 μ A (30 Hz);
 - Typical sleep mode power consumption: 2 μ A;
 - HBM $\geq$ 6000V, CDM $\geq$ 2000V;

- Latch up $\geq$ 200mA。
- Supports capacitive touch screens
 - Supports up to 20 drive/induction channels, with TX/RX switching capability and support for up to 10 RX channels;
 - It is recommended that the touchscreen size be $\leq$ 2.2 inches;
 - Supports open-channel or pull-down design;
 - Supports traditional DITO and SITO formats as well as various patterns, including G/FF, G/G, G/F, OGS, On-cell, etc.;
 - Automatic module parameter tuning, with maximum supported impedance of 20 k Ω ;
 - Cover Lens thickness specifications: glass $\leq$ 2 mm, acrylic $\leq$ 1 mm.
- communication interface
 - I2C master-slave communication interface with a configurable baud rate ranging from 50 kHz to 400 kHz;
 - Supports GPIO; offers multiple configurable operating modes; includes built-in pull-up resistor mode;
 - Equipped with a built-in 1.8V LDO; compatible with 1.8V/VDDA interface voltage levels, configurable as needed.
- Power Supply
 - Single power supply: 2.8–3.6 V; refer to the circuit design; power supply ripple $\leq$ 50 mV;
 - A small number of peripheral components.
- Package Type
 - WLCSP 30ball: 2.15mm $\times$ 2.15mm $\times$ 0.46T。

3. Apply

Watches, smart bands, scanning pens, etc.

4. Typical Application Circuit Diagram



C1: 2.2uF/10V

C2: 0.22uF/10V

C3: 0.22uF/10V, 当VBUS接HOST才需要, 其他情况可以去掉

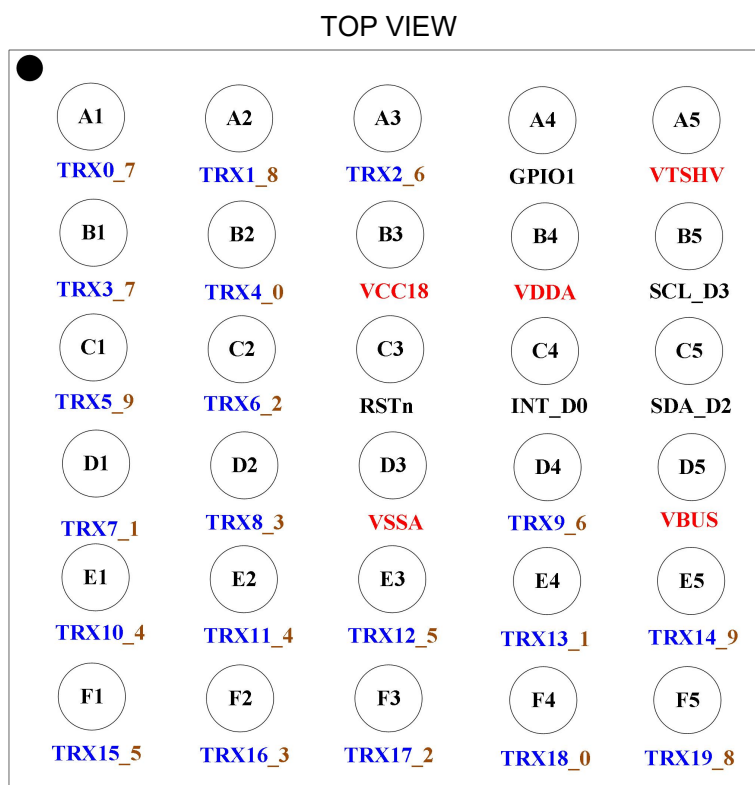
C4: 0.22uF/10V

VBUS: 必选, 建议HOST端提供, 根据IIC通讯电压决定

R1/R2: 选配, I2C 总线上拉电阻, 可配置芯片内部5K上拉电阻代替

TRX_m和TRX_n: 作为RX的n数字不能重复, 做为TX的m尽量不重复

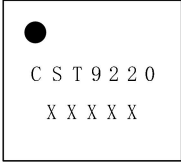
5. Pin Layout



6. Pin Description

Pin	Name	Type	Functional description	Pin	Name	Type	Functional description
A1	TRX0	I/O	Drive/Sensor Channel	D1	TRX7	I/O	Drive/Sensor Channel
A2	TRX1	I/O	Drive/Sensor Channel	D2	TRX8	I/O	Drive/Sensor Channel
A3	TRX2	I/O	Drive/Sensor Channel	D3	VSSA	GND	Analog ground
A4	GPIO1	I/O	General-purpose Digital I/O	D4	TRX9	I/O	Drive/Sensor Channel
A5	VTSHV	PWR/O	PUMP 8V, 0.22uF	D5	VBUS	PWR/I	Communication voltage
B1	TRX3	I/O	Drive/Sensor Channel	E1	TRX10	I/O	Drive/Sensor Channel
B2	TRX4	I/O	Drive/Sensor Channel	E2	TRX11	I/O	Drive/Sensor Channel
B3	VCC18	PWR/O	1.8V, 0.22uF	E3	TRX12	I/O	Drive/Sensor Channel
B4	VDDA	PWR/I	2.8~3.6V, 2.2uF	E4	TRX13	I/O	Drive/Sensor Channel
B5	SCL_D3	I/O	I2C Clock Signal	E5	TRX14	I/O	Drive/Sensor Channel
C1	TRX5	I/O	Drive/Sensor Channel	F1	TRX15	I/O	Drive/Sensor Channel
C2	TRX6	I/O	Drive/Sensor Channel	F2	TRX16	I/O	Drive/Sensor Channel
C3	RSTn	I	Reset; low effective.	F3	TRX17	I/O	Drive/Sensor Channel
C4	INT_D0	I/O	General-purpose Digital I/O	F4	TRX18	I/O	Drive/Sensor Channel
C5	SDA_D2	I/O	I2C Data Signal	F5	TRX19	I/O	Drive/Sensor Channel

7. Order Information

P / N	Package	Surface printing	Pack	Explain
CST9220	WLCSP 30ball 2.15×2.15×0.46T		4,000 rolls of cable-wound products shipped	Dot: Pin1 Mark CST9220: Model Number XXXXX: Production Tracking Code

8. Extreme Parameter Table

Parameter	Symbol	Least value	Representative value	Crest value	Unit	Explanatory note
Operating voltage VDDA	Vdd	2.8	3.0	3.6	V	Opposite VSSA
Power supply ripple	Vrip	-	-	50	mV	
Analog I/O voltage tolerance	Vioa	-0.3	-	8	V	
Digital I/O voltage tolerance	Viod	-0.3	-	3.6	V	
I/O maximum current rating	Iiom	-10	-	10	mA	
Operating temperature range	Topr	-20	+25	+85	°C	
Storage temperature range	Tstg	-60	-	+125	°C	
Work humidity	Hopr	-	-	95	%	
ESD HBM	ESD	±6000	-	-	V	Human Body Model ESD
ESD CDM	ESD	±2000	-	-	V	Charge Device Model
Latch-up Current	LU	-	-	200	mA	

9. Electrical Character

9.1 Direct Current (DC) Electrical Characteristics

Ambient temperature: 25 °C; VDDA = 2.8 V.

Parameter	Symbol	Least value	Representative value	Crest value	Unit
Low-level output voltage value	Vol	-	-	0.3*VDDIO	V
High-level output voltage value	Voh	0.7*VDDIO	-	-	V
Enter the low-level voltage value	Vil	-0.3	-	0.3*VDDIO	V
Enter high-level voltage value	Vih	0.7*VDDIO	-	VDDIO	V
Current (Dynamic Mode)	Iopr	-	1.4 (100Hz)	-	mA
Current (Monitoring Mode)	Imon	-	12 (30Hz)	-	uA

Current (Sleep Mode)	Islp	-	2	-	uA
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9.2 Alternating Current (AC) Electrical Characteristics

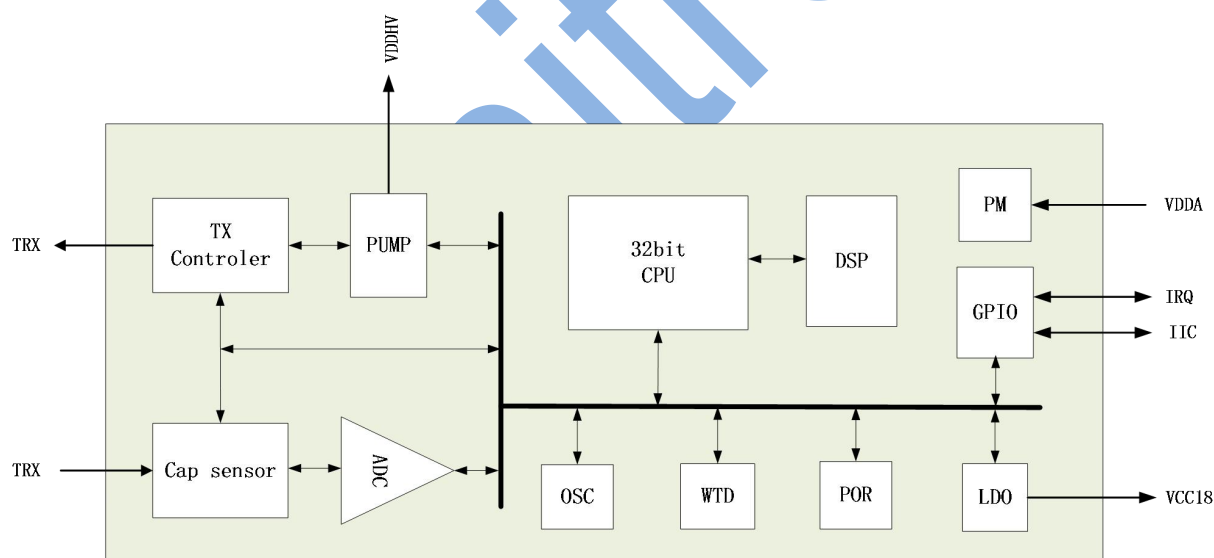
Ambient temperature: 25 °C; VDDA = 2.8 V.

Parameter	Symbol	Least value	Representative value	Crest value	Unit
TX clock frequency	ftx	-	-	300	kHz
TX output voltage	Vtx	-	-	7.5	V
RX input voltage	Vrx	-	1.4	-	V

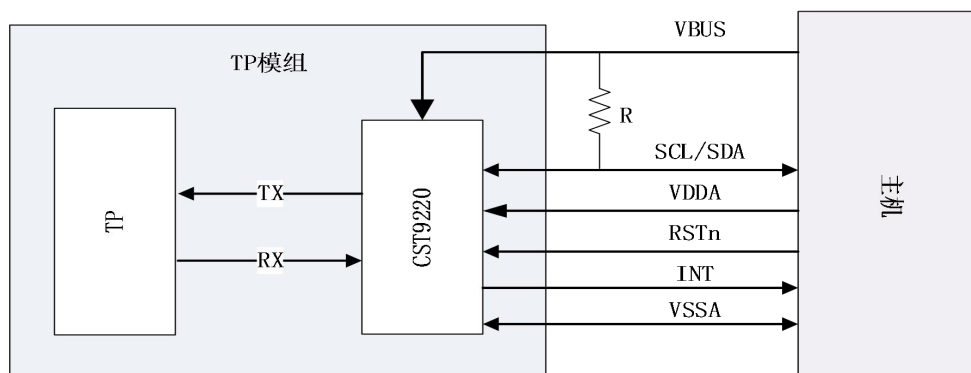
10. Functional Description

The CST9220 series multi-point capacitive touch ICs utilize high-voltage driving technology, which offers a higher signal-to-noise ratio and superior noise immunity compared to traditional low-voltage driving schemes, enabling ultra-sensitive touch detection. Additionally, the integrated self-interdigital capacitive sensing module, combined with an intelligent scanning algorithm, ensures rapid response times while delivering exceptional noise immunity, waterproofing, and low power consumption performance.

The overall system block diagram is as follows:



10.1 Host Interface



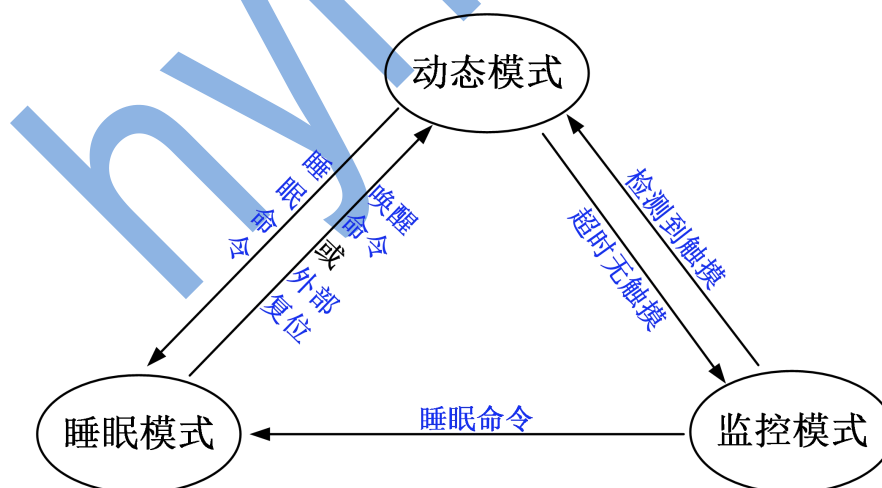
The figure above illustrates the interface relationship between the host and the CST9220: the interface between the host and the CST9220 includes the VBUS (optional), I2C, INT, RSTn, VSSA, and VDDA signals; the interface between the CST9220 and the TP includes the TX and RX signals.

VDDA: Operating voltage of the CST9220.

SCL and SDA: Serial communication interface; the host acts as the Master, and the CST9220 acts as the Slave.

INT: Interrupt signal – a universal GPIO interface used to notify the host that data is ready for reading when the CST9220 has prepared the data; examples of such data include touch data, gesture data, etc.

10.2 Work Pattern



- Dynamic mode: Upon chip power-on or reset, as well as during frequent touch events.

In this mode, the touch sensor chip rapidly performs an intelligent scan of the touchscreen, promptly detects touches, and reports them to the host. This mode offers the fastest response time, the best performance, and the highest power consumption.

- Monitoring Mode: Screen-off monitoring with touch-triggered wake-up; when the screen is on for a period of time but no touch input is detected.

In this mode, the touch sensor chip detects touch events at a relatively low scan frequency; once a touch event is detected, it

rapidly switches to dynamic mode. By combining this mode with DSP-based self-contained scanning, the system can quickly detect touch events and switch to dynamic mode. This mode enables the detection of touch events with extremely low power consumption.

- **Sleep Mode:** Turns off the screen without requiring any touch during active use.

Upon receiving a sleep command, the touch sensor chip enters deep sleep mode to maximize power consumption savings; it can be awakened via a wake-up command or an external reset method.

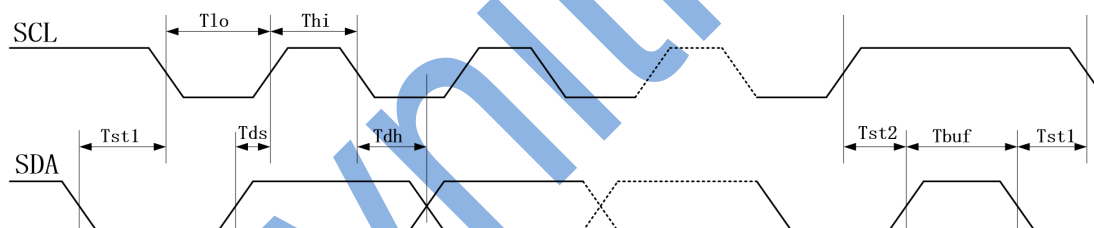
10.3 Channel/Node Configuration

The CST9220 multi-touch controller chip can support up to 17 channels, with each channel offering flexible configuration between drive and sensing functions; additionally, each channel supports self-interdigitated capacitive scanning. When assigning drive/sensing pins, it is recommended to select pins that are as continuously distributed as possible.

Supported mutual capacitance range per node: 0.5 pF – 20 pF (assuming a drive voltage of 7 V).

10.4 I2C Communication

The CST9220 supports the standard I2C communication protocol and can achieve configurable communication rates ranging from 50 kHz to 400 kHz. It features two I2C pins (SCL and SDA), which not only support open-drain mode but also support an internal pull-up configuration, providing flexible selection options.



Description	Symbol	Fast Mode		Unit
		Min	Max	
SCL clock frequency	Fscl	0	400	kHz
SCL hold time for START condition	Tst1	0.6	-	us
LOW period of SCL	Tlo	1.3	-	us
HIGH period of SCL	Thi	0.6	-	us
SDA setup time	Tds	0.1	-	us
SDA hold time	Tdh	0	0.9	us
SCL setup time for STOP condition	Tst2	0.6	-	us
Ready time between STOP and START	Tbuf	20	-	us

The CST9220 always acts as a slave device; the initialization process is initiated by the master device. During the period when the clock line SCL remains at a high level, the voltage on the data line SDA is pulled low (i.e., a negative transition), which is defined as the start signal for the I2C bus.

After the CST9220 detects the bus start signal, it sends an 8-bit address (which can be customized within the chip; the default value is 0xB4/0xB5). In the 9th clock cycle, the data line SDA is switched to an output port and pulled low to serve as the acknowledgment signal. The data line SDA then serially transmits 9 bits of data over 9 clock cycles, consisting of 8 bits of valid data plus 1 bit of the receiver's acknowledgment signal (ACK) or non-acknowledgment signal (NACK).

The stop signal is also actively generated by the host once communication has concluded. The stop signal occurs when the clock line SCL remains at a high level, while the data line SDA is released, causing SDA to return to a high level (i.e., a positive transition); this signal marks the termination of a data transfer.

a. The host writes data to the CST9220. The data transmission format is shown in the figure:

S	Slave Address[7bit]	W[1bit]	ACK	DATA[8bit]	ACK	...	DATA[8bit]	ACK/ NACK	P
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B. The host reads data from the CST9220. The data transmission format is shown in the figure:

S	Slave Address[7bit]	R[1bit]	ACK	DATA[8bit]	ACK	...	DATA[8bit]	NACK	P
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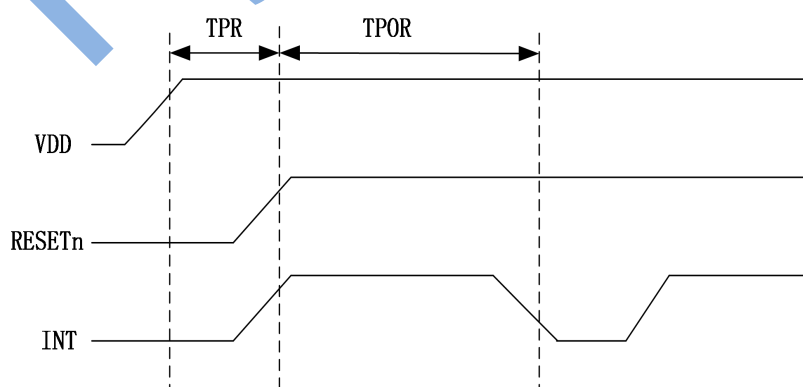
c. The host writes data to the CST9220, then resets the start condition, and subsequently reads data from the CST9220; alternatively, the host device reads data from the CST9220, then resets the start condition, and subsequently writes data to the CST9220. The data transmission format is shown in the figure:

S	Slave Address[7bit]	W[1bit]	ACK	DATA[8bit]	ACK	...	DATA[8bit]	ACK/ NACK	
RS	Slave Address[7bit]	R[1bit]	ACK	DATA[8bit]	ACK	...	DATA[8bit]	NACK	P

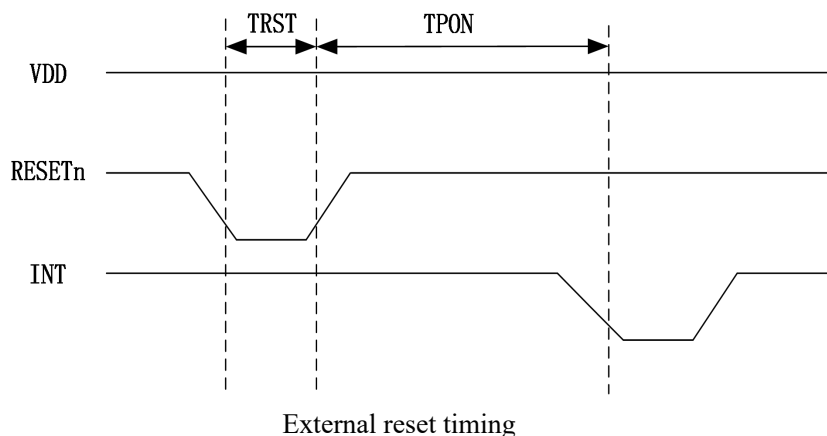
10.5 Power On/Reset

The built-in power-on reset module keeps the chip in a reset state until the voltage returns to normal; when the voltage falls below a certain threshold, the chip will also be reset. When the external reset pin (RSTn) is at a low logic level, the entire chip is reset; this pin incorporates an internal pull-up resistor and an RC filter, though the pin can also be left floating externally. The chip features a built-in watchdog timer that ensures the chip can return to normal operating mode within a specified time frame even in the event of abnormal conditions.

The power-on reset timing sequence is shown in the figure below:



Power-on Timing Sequence

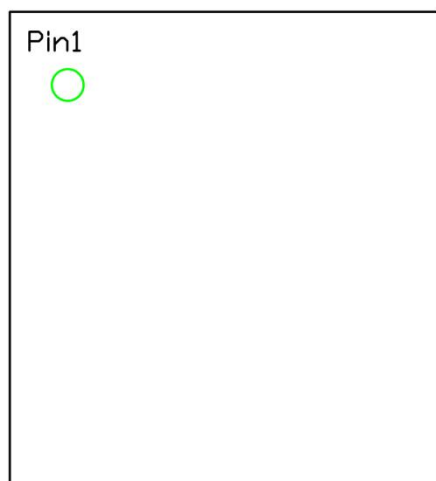


Symbol	Description	Representative value	Unit
TPOR	Chip initialization time upon power-on	100	ms
TPR	RST pin rising edge delay time	1	ms
TRON	Chip reinitialization time after reset	100	ms
TRST	Reset pulse duration	0.1	ms

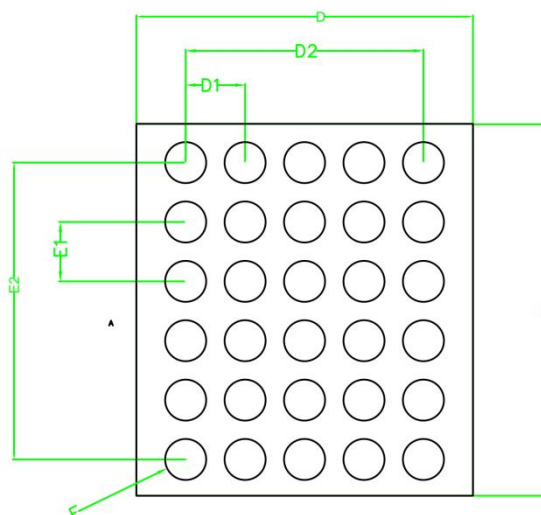
10.6 Interrupt Mode

The touch sensor chip will only notify the host to read valid data via the INT pin when it detects a valid touch and needs to report this event to the host; this approach improves efficiency and reduces the CPU load. The interrupt edge can be configured as either a rising or falling edge trigger, depending on requirements.

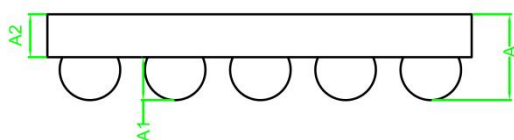
11. Product Packaging



Top View(Marking)



Bottom View(Ball up)



Side View

Item	No.	Mean/mm	Tolerance
总厚度	A	0.461	±0.0325
球高	A1	0.151	±0.020
Wafer Thickness	A2	0.310	±0.025
芯片尺寸	X	D	2.155 ±0.025
	Y	E	2.155 ±0.025
球尺寸	F	0.217	±0.020
球间距	D1	0.400	NA
	D2	1.600	NA
	E1	0.350	NA
	E2	1.750	NA

12. Colophon

Document Version	Revise	Time
V1.0	Initial Version	2022-10-28

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